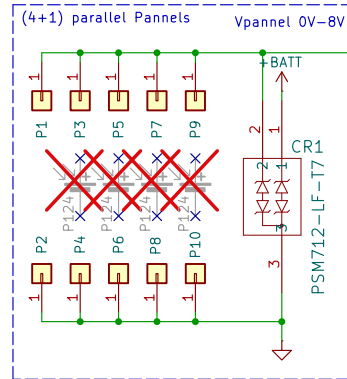


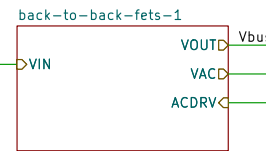
USB



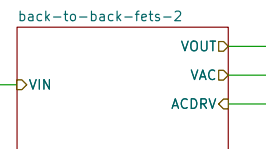
File: usb.kicad_sch



Allow supply selection and prevent back current

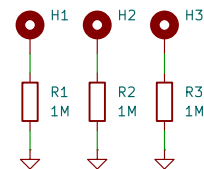


File: back-to-back-fets.kicad_sch

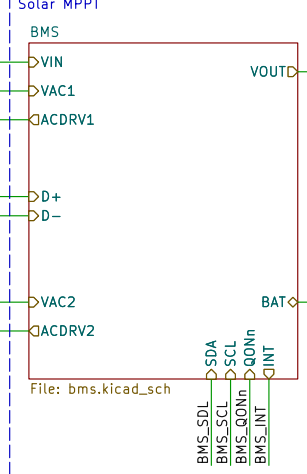


File: back-to-back-fets.kicad_sch

Mounting Holes

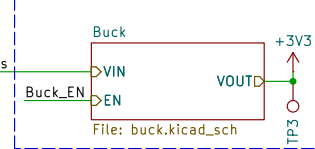


Charge / Discharge Battery Solar MPPT



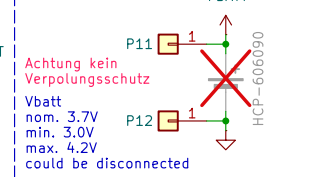
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Lower Battery Voltage to 3.3V

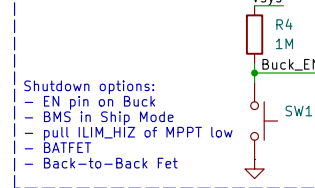


File: buck.kicad_sch

Battery

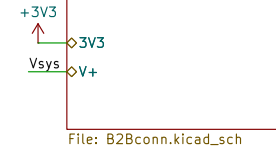


Shutdown

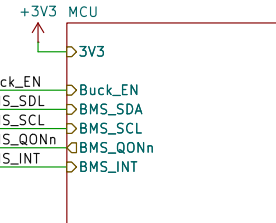


Shutdown options:
- EN pin on Buck
- BMS in Ship Mode
- pull ILIM_HIZ of MPPT low
- BATFET
- Back-to-Back Fet

BoardToBoardConnectors



File: B2Bconn.kicad_sch



File: mcu.kicad_sch

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Sheet: /

File: Amadee.kicad_sch

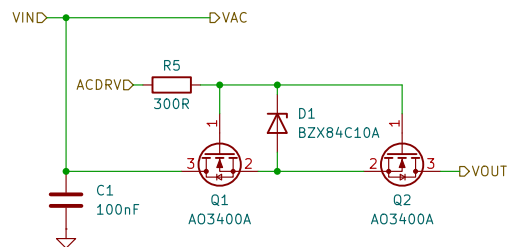
Title: Amadee Tumber

Size: A4 Date: 2026-05-13

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Rev: 0.1

Id: 1/8



Loss on both FETs at Solar MPP with FET@125°C -> $R_{ds(on)}=33mR$
 $(2 \cdot 0.033R \cdot (2 \cdot 0.18A)^2) / (2 \cdot 5.89V \cdot 0.18A) = 0.41\%$
 Could select FET with lower $R_{ds(on)}$,
 but AO3400A is a basic part so cheaper.

The zener limits the GS Voltage.
 max. for AO3400A is 12V
 leave 2V space so 10V Zener with low leakage @ 5V.

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Sheet: /back-to-back-fets-1/

File: back-to-back-fets.kicad_sch

Title: Amadee Tumber

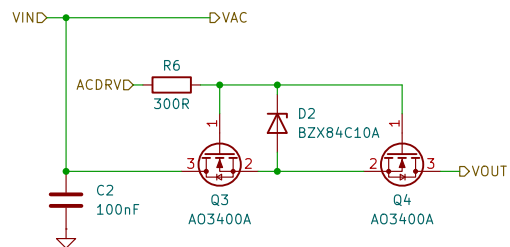
Size: A4

Date: 2026-05-13

Rev: 0.1

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Id: 2/8



Loss on both FETs at Solar MPP with FET@125°C -> $R_{ds(on)}=33mR$
 $(2*0.033R * (2*0.18A)^2)/(2*5.89V*0.18A) = 0.41\%$
 Could select FET with lower $R_{ds(on)}$,
 but AO3400A is a basic part so cheaper.

The zener limits the GS Voltage.
 max. for AO3400A is 12V
 leave 2V space so 10V Zener with low leakage @ 5V.

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Sheet: /back-to-back-fets-2/

File: back-to-back-fets.kicad_sch

Title: Amadee Tumber

Size: A4

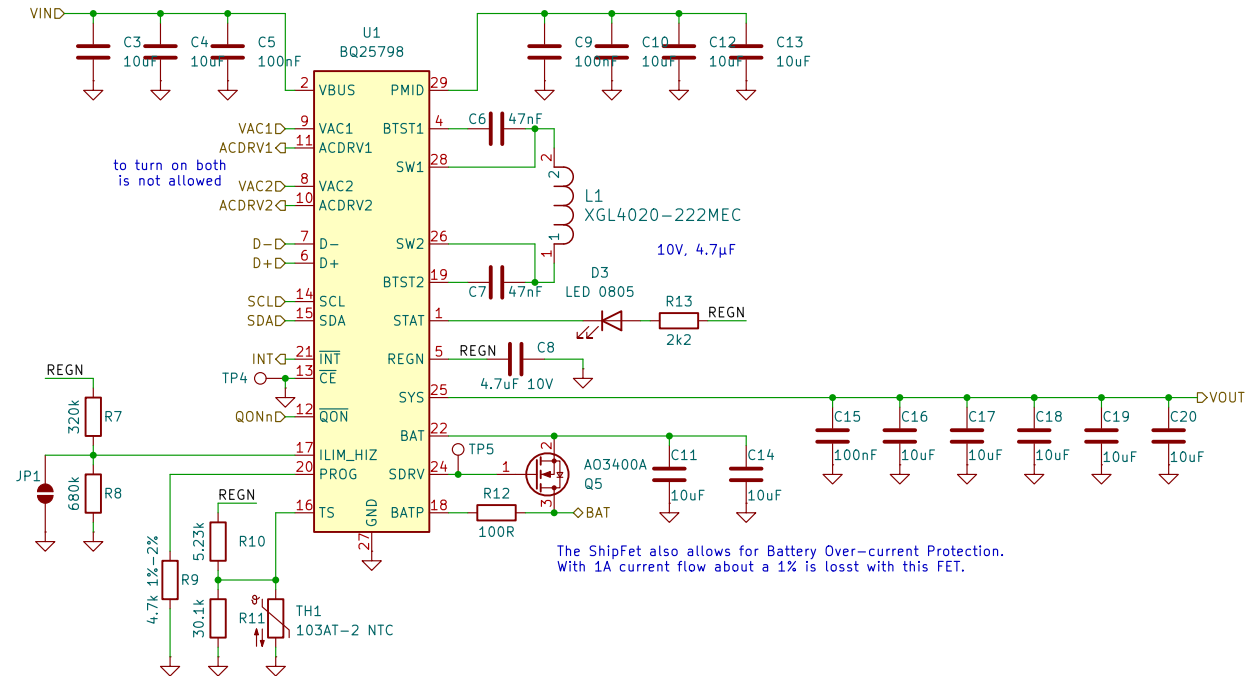
Date: 2026-05-13

Rev: 0.1

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Id: 3/8

BQ25798
MPPT
Efficiency for conversion to 1s Battery Charge better 90%
BATFET 11mR
Battery Lade/Entlade effizienz?



750kHz switching frequency to achieve higher efficiency
-> downside requires larger inductor (2.2 μ H) and capacitors.
set by 4k7 on PROG-Pin for 1S Lipo see, Datasheet

max input current limit:

- Solar cells will have MPPT
- USB will detect limit with D+ and D-
- When charging on a Powersupply the Input Current Optimizer will find the best input current
- The battery charging current should be limited by the REG03_Charge_Current_Limit Register
- The back to back fets allow over 4A

We can set a limit on the REG06_Input_Current_Limit Register with a voltage on VILIM_HIZ.

$VILIM_HIZ = 1V + 800m\Omega \times ILIM$

I don't really see any reason to limit the input current.

The Datasheet gives a maximum of 3.3A input current so let's leave some margin and go for 3A.

$1V + 800m\Omega \times 3A = 3.4V$

Voltage divider from REGN = 5V

$680k \times (5V/3.4V - 1) = 320k$

The ILIM_HIZ-Pin might leak +- 1.5uA

(Connect ILIM_HIZ to REGN to set the maximum input current limit.)
(could this pin be used to put the IC in High Impedance mode?)

Select T1=0°C and T5=60°C for Li-ion or Li-polymer battery, the RT1 and RT2 are calculated to be 5.24K Ω and 30.31K Ω respectively.

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Sheet: /BMS/

File: bms.kicad_sch

Title: Amadee Tumber

Size: A4 Date: 2026-05-13

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Rev: 0.1

Id: 4/8

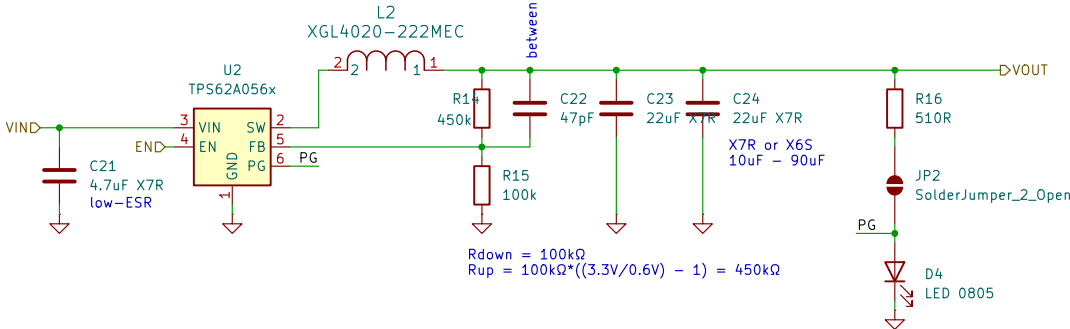
TPS62A056x
2mA to 1A better than 95%eff for 4.2V to 3.3V

In 100% Duty Cycle Low Dropout Operation:

$$V_{out} = V_{in} - I_{out}(R_{ds(on)} + R_L)$$

$$= V_{in} - 250mA(100m\Omega + 22m\Omega)$$

$$= V_{in} - 30.5mV$$
 loss @ 3.0V 250mA ~ 1%
 loss @ 3.0V 25mA ~ 0.1%



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Sheet: /Buck/

File: buck.kicad_sch

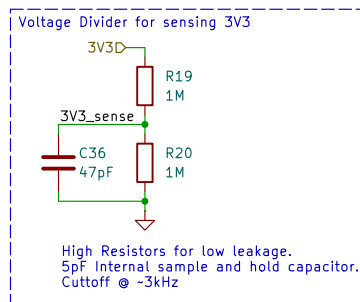
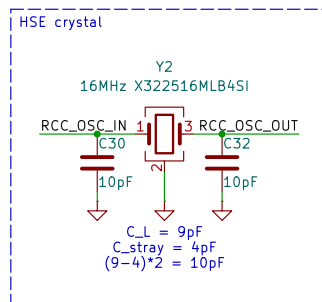
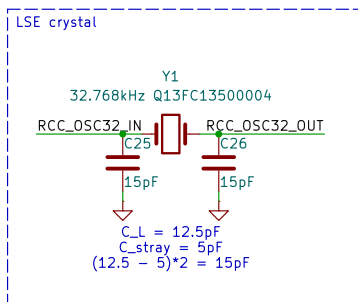
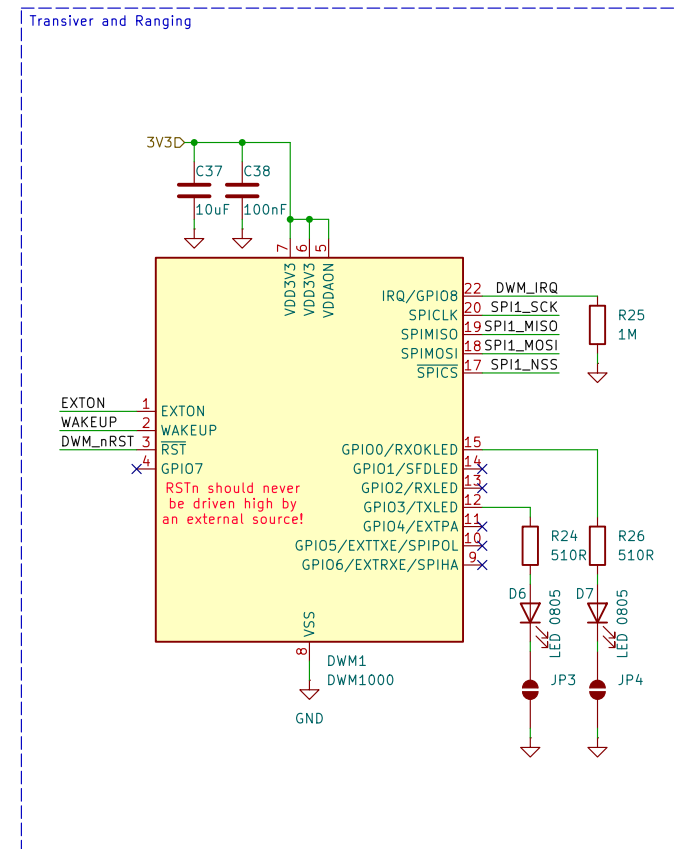
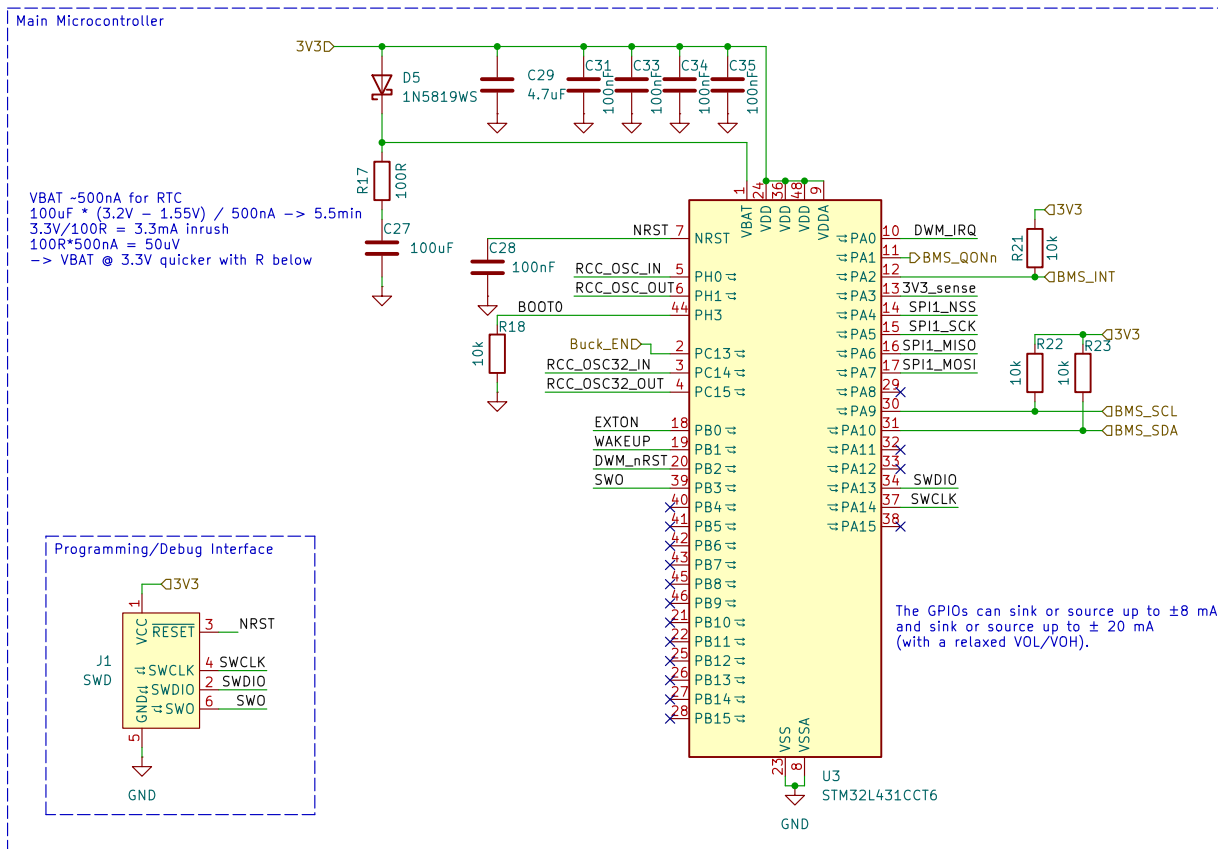
Title: Amadee Tumber

Size: A4	Date: 2026-05-13
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Rev: 0.1

Id: 5/8



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 Sheet: /MCU/
 File: mcu.kicad_sch

Title: Amadee Tumber

Size: A4 Date: 2026-05-13
 KiCad E.D.A. 10.0.2

Rev: 0.1
 Id: 6/8

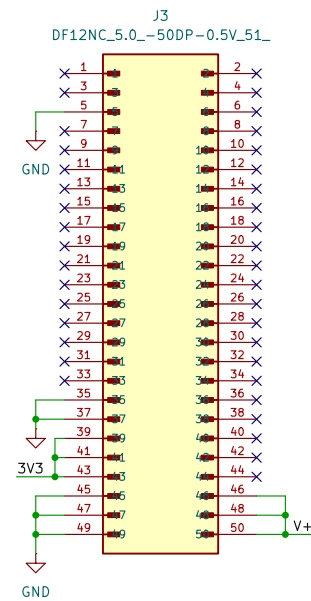
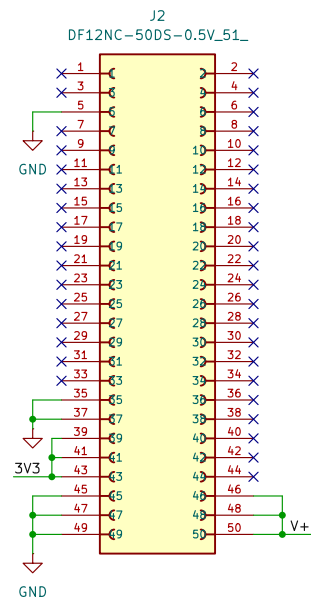
This is intended to power the skith board

JP5
SolderJumper_2_Open
3V3

JP6
SolderJumper_2_Open
V+

H4
MountingHole

H5
MountingHole



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Sheet: /BoardToBoardConnectors/
File: B2Bconn.kicad_sch

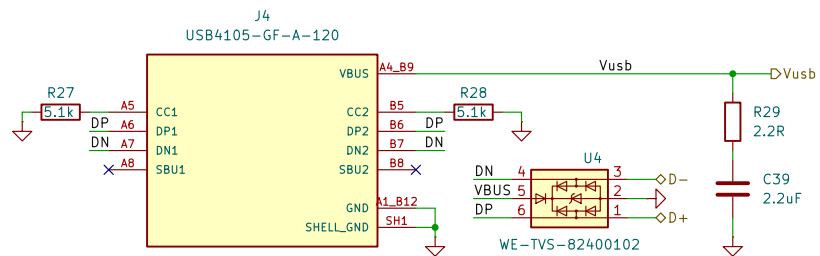
Title: Amadee Tumber

Size: A4 Date: 2026-05-13

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Rev: 0.1

Id: 7/8



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Sheet: /USB/

File: usb.kicad_sch

Title: Amadee Tumber

Size: A4

Date: 2026-05-13

Rev: 0.1

KiCad E.D.A. 10.0.2

Id: 8/8